

Data Formats: XML & JSON

Week 4 · Foundations module · Textbook Chapter 4

Brian C. Keegan, Ph.D.

Associate Professor, Department of Information Science
University of Colorado Boulder

September 9 & 11, 2026

Two formats carry almost all web data: **XML** (a tree of tags) and **JSON** (nested lists and dictionaries). This week we learn to parse both.

Monday | Labor Day — no class

- Monday's concept intro is **folded into Wednesday**

Wednesday | Concepts & notebook lab

- XML & JSON concepts, then the Chapter 4 notebook: Congress, tweets, and weather

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 4

Companion notebook

`ch-04-data-formats.ipynb`

Short week

No class Monday (Labor Day). We meet **Wednesday** (concepts + notebook) and **Friday** (revisions).

1. Monday • Concepts

Why data formats matter — No class Monday (Labor Day) — these concepts open Wednesday's session, just before the notebook.

Before you can *extract* data from the web, you have to understand how it is *structured*. Two formats dominate:

- **XML** — the older, document-centric format; a tree of nested tags. Born from the 1990s web.
- **JSON** — the newer, lighter format of the API economy; maps directly onto Python lists and dictionaries.

Every page, API response, and data feed you meet this semester is one of these — or HTML, which is XML's sibling.

When to use which

XML: older web services, government feeds (Congress, SEC, RSS), configuration files.

JSON: modern web APIs, database exports, service-to-service exchange.

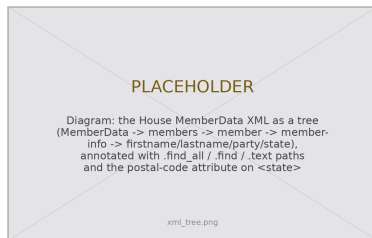
XML: a tree of nested tags

XML organizes data as a **hierarchy**: each tag can hold text, attributes, and child tags.

XML and HTML are **siblings**, not parent and child — both descend from SGML. Only the stricter XHTML dialect is well-formed XML; the HTML in the wild is not, which is exactly why the forgiving parsers of Ch. 6 exist.

A value can hide in one of two places:

- **text** — between an opening and closing tag (read with `.text`)
- **attributes** — key/value pairs inside the opening tag (read with brackets)



The House `MemberData` a feed, drawn as a tree.

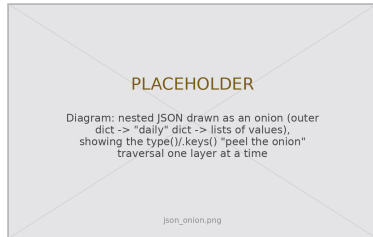
JSON: lists and dictionaries you already know

JSON maps onto two Python structures you already use:

- a JSON **array** is a Python **list** — ordered, indexed by position
- a JSON **object** is a Python **dict** — accessed by key

Real data **nests** them — dicts inside lists inside dicts, several levels deep. The skill is navigating from the outside in, one key or index at a time.

Created by Douglas Crockford in the early 2000s; the rise of AJAX made JSON the default API format by the 2010s.



“Peel the onion”: `type()`, then `.keys()`, one layer deeper each time.

One pipeline unifies the whole course

Whether data starts as XML, JSON, HTML, or PDF text, the move into analysis is always the same:

find records → **extract fields** →
list of dictionaries → `pd.DataFrame`

The DataFrame is the **universal intermediate format**. Reach it and all of pandas opens up: filter, group, cross-tabulate, merge.

The source changes; the tag names change; **the pipeline stays the same**.

You reuse this pattern in...

- HTML tables — Ch. 6
- archived pages — Ch. 7
- PDF text — Ch. 9
- Wikipedia, government & social APIs — Ch. 10–12

2. Wednesday · Notebook Lab

Parse the string into a navigable tree, then search it. `.find()` returns the first match, `.find_all()` returns them all — each searches the *entire* subtree beneath a tag:

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(house_raw, "xml") # the "xml" parser comes from lxml

members = soup.find_all("member")
print(len(members)) # 2

for m in members:
    first = m.find("firstname").text # .text = content between the tags
    last = m.find("lastname").text # .find() reaches nested tags too
    party = m.find("party").text
    print(f"{first} {last} ({party})") # Joe Neguse (D) / Jeff Hurd (R)
```

The two-place problem — and the crash that guards it

A value is either **text** or an **attribute**:

```
state = soup.find("state")
state["postal-code"] # "CO"
state.find("state-fullname").text # "Colorado"
```

A missing tag makes `.find()` return `None`:

```
# AttributeError: None has no .text
m.find("party").text
# guard every extraction instead:
m.find("party").text if m.find("party") else None
```

Working out *which* of the two holds your value is half the parsing battle — real XML uses both.

Calling `.text` on a `None` is the **single most common** scraping error. The `if ...else None` guard keeps one missing field from crashing a 441-record loop.

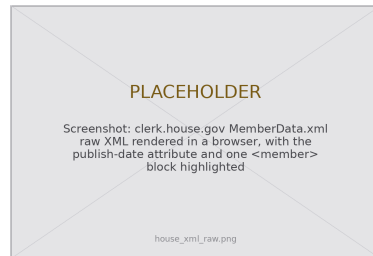
XML always hands you **text**: converting "2nd" or "\$1,234" to a number is your job.

XML → DataFrame: the real House roster

```
import requests, pandas as pd

url = "https://clerk.house.gov/xml/lists/MemberData.xml"
soup = BeautifulSoup(requests.get(url).text, "xml")
print(soup.find("MemberData")["publish-date"])

records = []
for m in soup.find_all("member"): # 441 members
    st = m.find("state")
    records.append({"last_name": m.find("lastname").text,
                    "party": m.find("party").text,
                    "state": st["postal-code"] if st else None})
df = pd.DataFrame(records) # find_all -> dicts -> DataFrame
```



The Clerk's live MemberData.xml.

JSON: parse a string, then the list-of-dictionaries pipeline

`json.loads()` turns a string into Python objects (`json.load()` reads a file). With `requests`, `response.json()` does it for you:

```
import json, pandas as pd

data = json.loads('{"state": "Colorado", "population": 5773714}')
print(data["state"]) # Colorado
# response.json() == json.loads(response.text)

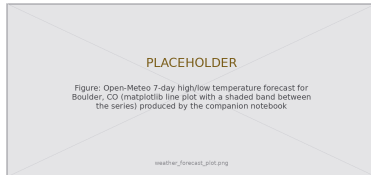
census = [ # the list-of-dictionaries pattern:
    {"state": "Colorado", "year": 2020, "population": 5773714},
    {"state": "Colorado", "year": 2010, "population": 5029196},
]
df = pd.DataFrame(census) # one dict per row, keys -> columns
```

JSON: peel the onion on a live API

```
import requests

url = "https://api.open-meteo.com/v1/forecast"
params = {"latitude": 40.01, "longitude": -105.27,
          "daily": "temperature_2m_max",
          "timezone": "America/Denver"}
data = requests.get(url, params=params).json()

# type() + .keys(), one level deeper each time
data.keys() # latitude, longitude, daily...
data["daily"].keys() # time, temperature_2m_max
highs = data["daily"]["temperature_2m_max"] # a list
data.get("elevation", "unknown") # .get avoids KeyError
```



Boulder's 7-day forecast, straight from the nested lists.

Lab: work these in pairs, then show-and-tell

Work in pairs. Do these exercises from the Chapter 4 notebook:

1. **RSS feed** — parse a news feed with the "xml" parser. Extract the title, the link, and the date. Put them in a DataFrame.
2. **Nested JSON** — get the dates and the maximum temperatures from Open-Meteo. Put them in a DataFrame.
3. **XML and JSON** — get the same data in both formats. Report which format is easier to parse. Report which format has more metadata.
4. **Reusable** — write `xml_to_dataframe(soup, tag, fields)`. Test it on the Congress data. Then test it on an RSS feed.
5. **Weather** — plot the high and low temperatures. Mark the days below freezing.
6. **5617**: write a critique of a public XML or JSON dataset. Use the datasheet format of Gebru et al. (2021)

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

The data source changes.

The tag names change.

The pipeline stays the same.

`find_all` → `dicts` → `DataFrame`.

3. Friday • Textbook Revisions

Friday — improving Chapter 4 together

Today we make the book better. Each of you opens a **pull request** against `ch-04-data-formats.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable



High-value targets — pick one and scope it to a single PR:

- **Test the code against live sources.** The House roster drifts (441 today; the `publish-date` changes) and Open-Meteo dates roll forward. Run the notebook, report what changed, and pin the snapshot.
- **Close a gap in “Common Issues to Debug.”** Add a worked `json.JSONDecodeError` case — a trailing comma, single quotes, or an HTML error page returned instead of JSON.
- **Add a figure.** The chapter has *no* diagrams: contribute a clean XML-tree or a JSON “peel-the-onion” figure.
- **Strengthen an exercise.** Give the RSS or weather exercise expected output, a rubric, or a starter cell so it is self-checking.
- **Extend the RSS / post-API story.** Tie the decline-of-RSS narrative to a concrete citation and the “soft enclosure” theme (Ch. 3).

What makes a good revision & a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Renders (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates “must fix” from “nice to have”
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Web Architecture & Protocols** — how a URL becomes a rendered page (HTTP, requests, status codes).

Key takeaways

1. **XML** is a tree of tags; **JSON** is nested lists and dictionaries — the two structural languages of web data.
2. BeautifulSoup navigates XML with `.find()`, `.find_all()`, and `.text`; attributes come out with bracket notation.
3. For JSON, **peel the onion** — inspect `type()` and `.keys()` top-down, and use `.get()` for safe access.
4. The `find_all` → list-of-dicts → `DataFrame` pipeline is the bridge from raw web data to analysis.
5. Format is a **transparency** choice: open XML/JSON serves public oversight; scanned PDFs quietly enclose it.